Example

Minimize $0.5x + 3x + 4y$

$\frac{1}{2}[x \quad 1 \quad x] + y \quad x$

Subject to

$x \leq 0$

$y \leq 0$

$x + 3y \leq 15$

$2x + 3y \geq 100$

$3x + 4y \geq 80$